

HealthPulse

A Full-Stack MERN Health & Wellness Platform

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Problem Statement

- The internet is saturated with **fragmented** and sensationalized health advice.
- Traditional CMS platforms (like WordPress) are excessively **bloated and slow**.
- Legacy systems rely on heavy plugins, creating severe **security vulnerabilities**.
- Non-technical publishers face **steep learning curves** for simple content management tasks.



Our Solution

- Bespoke SPA:** A lightweight, decoupled Single Page Application engineered for maximum speed and readability.
- Real-Time Grid:** A responsive, public-facing article grid that filters content instantaneously without page reloads.
- Secure Dashboard:** An intuitive, authenticated administrative portal allowing authors to manage content securely from anywhere.

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Your daily dose of wellness, nutrition, and mental health tips.

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The MERN Technology Stack



Frontend

Built with **React.js** and scaffolded via **Vite** for lightning-fast module replacement. Utilizes **React-Router** for seamless client-side navigation.



Backend API

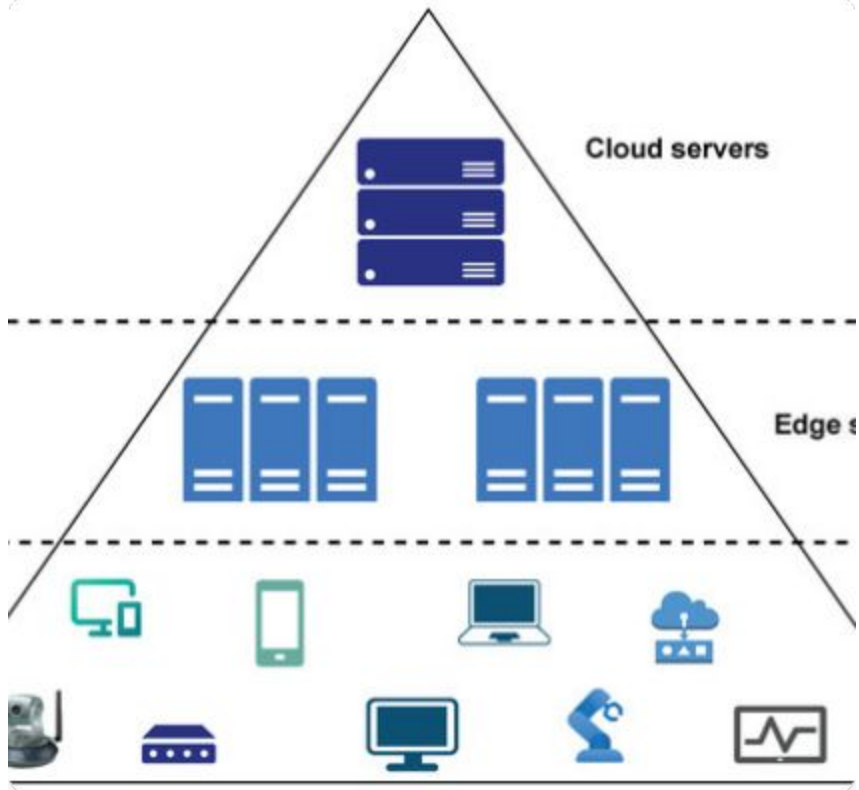
Powered by the **Node.js** runtime environment. We implemented **Express.js** to handle secure, stateless RESTful API routing and **Axios** for HTTP requests.



Database

Data persistence achieved via **MongoDB Atlas Cloud**. We utilized the **Mongoose** Object Data Modeling (ODM) library for strict schema validation.

System Architecture (3-Tier Flow)



- **Presentation Tier (React/Vite):** Manages DOM rendering, local UI state, and captures user inputs directly from the browser.
- **Logic Tier (Express API):** Operates on Port 5001. Acts as the critical middleman, authenticating requests and processing business logic.
- **Data Tier (MongoDB Atlas):** Secure cloud cluster that stores flexible, JSON-like BSON documents across collections.

Key Features Developed



Real-Time Filtering: Instantaneous client-side search and category filtering using JavaScript array methods, eliminating database latency.



Protected Routes: Administrative dashboard secured via route-guarding logic and localized browser authentication (LocalStorage).



Full CRUD Operations: Complete graphical interface to Create, Read, Update, and Delete database documents.



Rich Text Editor: Integrated the react-quill-new WYSIWYG editor for professional article formatting (HTML injection).

Engineering Challenges & Debugging

Conquering CORS Blocks

Browsers initially blocked our frontend from sending POST payloads. We successfully diagnosed and defeated this Same-Origin Policy by strictly configuring Express **preflight headers** and explicitly authorizing the Vite client origin.

macOS Port Conflicts

We encountered persistent, silent connection failures during testing. Deep debugging revealed that the macOS **AirPlay Receiver** natively hijacks Port 5000. We re-architected our environment variables to bind the backend to Port 5001.

Conclusion & Future Scope



Cloud Image Hosting

Future iterations will migrate from external URL linking to direct multipart file uploads integrated with **AWS S3** or **Cloudinary** APIs.



Advanced Auth

Replacing local storage passwords with industry-standard **JSON Web Tokens (JWT)** and Bcrypt hashing to support multiple author roles.



Interactive Community

Developing a relational **comments schema** in MongoDB to allow public users to engage in active discussions beneath articles.